

Exponent Primality Does Not Imply Mersenne Primality

A pre-registered closed negative at $p = 11$, the verified Euclid–Euler perfect-number core, and the honest odd-perfect frontier

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github.com/dancinlab/hexa-lang

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Abstract

The MERSENNE axis of the TECS-L domain re-grounds the classical bridge between perfect numbers and Mersenne primes — the Euclid–Euler theorem, the divisor structure $\tau(2^{p-1}M_p) = 2p$, and the abundancy identity $\sigma(P) = 2P$ — as machine-checked closed-form recomputations under a single verification discipline (`hexa verify`, tier rubric §3). Against that verified backdrop we pre-register a single sharp falsifier and report one *closed-negative* headline finding. *Finding (MR6)*. We pre-registered Euclid’s implication that the converse of the Mersenne-prime hypothesis holds — that p prime forces $M_p = 2^p - 1$ to be prime. Testing the smallest non-trivial prime exponent past the classical list, $p = 11$, the recomputation returns $\sigma(2047) = 2160 \neq 2048$ and $\tau(2047) = 4 \neq 2$: $M_{11} = 2047$ is *composite*, with the exact factorization $2047 = 23 \times 89$ confirmed by $\sigma(23) = 24$, $\sigma(89) = 90$ (both witnessing primality). The falsifier is ∇ FALSIFIED: prime exponent does *not* imply Mersenne primality, so Euclid’s construction $2^{p-1}M_p$ does not produce a perfect number at every prime p — the Mersenne-prime hypothesis is *essential*, not decorative. Two further composite witnesses ($M_{23} = 47 \times 178481$, $M_{29} = 233 \times 1103 \times 2089$) confirm this is not a $p = 11$ accident. The ruled-out axis is the hypothesis that *exponent primality alone generates perfect numbers*. The positive backdrop — the Euclid–Euler equivalence and the abundancy/divisor closed forms across the first seven even perfect numbers, all $\blacklozenge$ — survives the negative intact. We are honest about the one residual: whether any *odd* perfect number exists is OPEN ($\bigcirc$), none below 10^{1500} , and we state it as a frontier, never as a closed result.

1 Introduction

A perfect number is a positive integer equal to the sum of its proper divisors; equivalently, $\sigma(n) = 2n$ where σ is the divisor sum. The first four were known to the Greeks: 6, 28, 496, 8128. Euclid (*Elements* IX.36, c. 300 BCE) gave a generating construction: if $2^p - 1$ is prime, then $2^{p-1}(2^p - 1)$ is perfect [1]. Two millennia later Euler (posthumous, 1849) proved the converse for the even case: *every* even perfect number arises this way [2]. Together these are the Euclid–Euler theorem: an even integer n is perfect if and only if $n = 2^{p-1}M_p$ for a Mersenne prime $M_p = 2^p - 1$ [5]. The numbers $M_p = 2^p - 1$ are the Mersenne numbers, and the search for Mersenne primes is today the province of the Great Internet Mersenne Prime Search [4].

The MERSENNE axis of the TECS-L domain deepens the perfect-number thread of an earlier $n = 6$ identity result [12] by re-grounding this entire classical edifice as machine-checked recomputation. Every perfect number, every abundancy value, every divisor count is reissued through a single verification command (`hexa verify`) and persisted verbatim, so that the claims of the paper are not assertions but reproducible verdicts [11].

A claim about Mersenne primes is scientific only to the extent that it comes with a *falsifier* fixed before the computation. The sharpest falsifier latent in Euclid’s construction is its tempting but *false* converse: does *every* prime exponent p yield a Mersenne prime M_p , and hence a perfect number? Euclid’s sufficiency direction needs M_p to be prime; if prime p already forced M_p prime, the hypothesis would be redundant. We pre-register exactly this implication as a falsifier and test it at the smallest prime exponent beyond the classical list whose Mersenne number is composite, $p = 11$.

The recomputation returns a deterministic negative. $M_{11} = 2047$ has $\sigma(2047) = 2160 \neq 2048$ and $\tau(2047) = 4 \neq 2$ — a composite number, with exact factorization $2047 = 23 \times 89$ (both 23 and 89 independently verified prime). Hence p prime does *not* imply M_p prime: Euclid’s construction fails to produce a perfect number at $p = 11$. The Mersenne-prime hypothesis is essential. This is a *closed-negative* finding — a pre-registered falsifier refuted by a real measurement, deterministically ruling out one axis — and in the TECS-L discipline such negatives are valuable precisely because they are closed [13].

The contribution of this paper is therefore: (i) a verified perfect-number / Mersenne backdrop — the Euclid–Euler equivalence, $\tau(2^{p-1}M_p) = 2p$, and $\sigma(P) = 2P$ across the first seven even perfect numbers, all $\blacklozenge$ (§3, §4); (ii) a single pre-registered falsifier resolved to a deterministic closed negative at $p = 11$, with exact factorization and two corroborating composite witnesses (§5); and (iii) an honest delineation of what the negative does and does not rule out, including the one genuinely open residual — the existence of odd perfect numbers (§6). Every numeric quantity traces to a verbatim `hexa verify` verdict under `.verdicts/tecs-l-mersenne-*/` and is indexed in `CLAIMS.tape` [11].

2 Background

Perfect numbers and the divisor sum. For a positive integer n , the divisor sum is $\sigma(n) = \sum_{d|n} d$ and the divisor count is $\tau(n) = \sum_{d|n} 1$. The number n is *perfect* when the sum of its proper divisors equals n , i.e. $\sigma(n) = 2n$, equivalently the abundancy ratio $\sigma(n)/n = 2$. The first five perfect numbers are 6, 28, 496, 8128, 33550336, computed by Euclid’s rule from the exponents $p = 2, 3, 5, 7, 13$ [5].

Mersenne numbers and primes. The p -th Mersenne number is $M_p = 2^p - 1$. A necessary condition for M_p to be prime is that the exponent p itself be prime: if $p = ab$ with $1 < a, b$, then $2^a - 1 \mid 2^p - 1$, so a composite exponent gives a composite Mersenne number. This necessary condition is the *forward* half. The seductive question — the subject of this paper — is whether it is also *sufficient*: does p prime force M_p prime? It does not, and the first witness is small.

The Euclid–Euler theorem. An even positive integer n is perfect if and only if $n = 2^{p-1}M_p$ where $M_p = 2^p - 1$ is a Mersenne prime [1, 2]. Euclid proved sufficiency (a Mersenne prime yields a perfect number); Euler proved necessity (every even perfect number is of this form). The theorem says nothing about *odd* perfect numbers, whose existence remains open (§6). The bridge is the spine of the MERSENNE axis: it ties the divisor-sum world (perfect numbers) to the prime-search world (Mersenne primes).

The Lucas–Lehmer test. Primality of M_p is decided by the Lucas–Lehmer recurrence: set $S_0 = 4$ and $S_{k+1} = S_k^2 - 2 \bmod M_p$; then M_p is prime if and only if $S_{p-2} \equiv 0 \pmod{M_p}$ [7]. For $p = 11$ the recurrence terminates at $S_9 \equiv 1736 \not\equiv 0 \pmod{2047}$, certifying M_{11} composite without exhibiting a factor. The factor witness $2047 = 23 \times 89$ is the complementary, constructive certificate this paper uses.

3 Method: the verification discipline

Every numeric claim in this paper is a closed-form recomputation issued through the `hexa verify` command and persisted verbatim. We use four tiers:

- **◆ formal** — a `hexa`-native closed-form recomputation that equals its expected value exactly (`hexa verify --expr <fn> <n> <v>` reporting `SUPPORTED-FORMAL`). These are the verified σ , τ , and `is_perfect` values.
- **■ numerical** — a `libm`-class value to a fixed tolerance, or a reasoned synthesis composed entirely from terminal **◆** atoms (the Euclid–Euler and abundancy syntheses). The MERSENNE axis is otherwise integer-exact; no floating-point tolerance is invoked.
- **▽ closed-negative** — a pre-registered falsifier that a recomputation deterministically refutes, ruling out one axis. This is the headline finding (MR6).
- **○ open / deferred** — a residual deliberately gated *out* of any terminal claim (the odd-perfect frontier, MR7), reported as an honest open problem and never used as a finding.

The single pre-registered falsifier (MR6).

Falsifier 1. Euclid’s converse holds: for every prime exponent p , the Mersenne number $M_p = 2^p - 1$ is prime — equivalently $\sigma(M_p) = M_p + 1$ and $\tau(M_p) = 2$ for all prime p — so that $2^{p-1}M_p$ is perfect at every prime p .

Method: recompute $\sigma(M_p)$ and $\tau(M_p)$ for the smallest prime exponent past the classical list whose Mersenne number is suspected composite, $p = 11$ ($M_p = 2047$). A composite witness is decisive: if $\tau(2047) \neq 2$ then 2047 has a non-trivial divisor and the falsifier is refuted. To make the refutation *constructive* we additionally verify the exact factorization $2047 = 23 \times 89$ by checking that both 23 and 89 are prime ($\tau = 2$, $\sigma = n + 1$). For robustness against a single-instance accident we repeat the composite test at the next two prime exponents whose Mersenne numbers are composite, $p = 23$ and $p = 29$.

The verified backdrop. The positive context against which the falsifier is sharpened is itself recomputed: `is_perfect` and the abundancy $\sigma(P) = 2P$ for the sixth and seventh perfect numbers ($P_6 = 8589869056$, $P_7 = 137438691328$, from the Mersenne-prime exponents $p = 17, 19$), and the closed-form divisor count $\tau(2^{p-1}M_p) = 2p$ across all seven perfect numbers $P_1, \dots, P_7$. The first five perfect numbers were already verified in the founding axis [12]; the MERSENNE axis extends the recomputation to P_6, P_7 and reframes the divisor structure in closed form.

The full driver is the HEAD `hexa verify --expr` engine. Raw standard output of every run is persisted verbatim under `.verdicts/tecs-1-mersenne-*/` and indexed in `CLAIMS.tape` [11]; representative transcripts are reproduced in Appendix E.

4 Verification: the real runs

4.1 The verified Euclid–Euler backdrop

The MERSENNE axis first recomputes the perfect-number / Mersenne backdrop, all **◆** except the two reasoned syntheses (**■**, composed from **◆** atoms). The headline counts (verbatim from the persisted verdicts) are:

backdrop result	sweep	tier count
<code>is_perfect</code> (P_6), <code>is_perfect</code> (P_7)	2 values	2/2 ◆
abundancy $\sigma(P_6) = 2P_6$, $\sigma(P_7) = 2P_7$	2 values	2/2 ◆
divisor count $\tau(2^{p-1}M_p) = 2p$ (P_1 – P_7)	7 values	7/7 ◆
Lucas–Lehmer cross-anchor (<code>is_perfect</code> for $p = 3, 5, 7, 13$)	4 values	4/4 ◆
Euclid–Euler synthesis (table over P_1 – P_7)	1 artifact	1 ■
abundancy closed-form $\sigma(P) = 2P$ synthesis	1 artifact	1 ■

The abundancy identity $\sigma(P) = 2P$ has an elementary closed-form derivation that the persisted synthesis records (Appendix D): since σ is multiplicative and $\gcd(2^{p-1}, M_p) = 1$,

$$\sigma(2^{p-1}M_p) = \sigma(2^{p-1})\sigma(M_p) = M_p \cdot 2^p = (2^p - 1)2^p = 2 \cdot 2^{p-1}M_p = 2P,$$

using $\sigma(2^{p-1}) = 2^p - 1 = M_p$ and $\sigma(M_p) = M_p + 1 = 2^p$ (the latter *requiring* M_p prime). The dependence on the Mersenne-prime hypothesis at the step $\sigma(M_p) = M_p + 1$ is exactly what Finding 1 probes: if M_p is *not* prime, $\sigma(M_p) > M_p + 1$ and P is no longer perfect. The seven $\sigma(P_k) = 2P_k$ instances (Appendix B) confirm the closed form numerically, the last two (P_6, P_7) verified here.

4.2 Finding run: M_{11} is composite (σ , τ witnesses)

Recomputing the divisor functions at the focal exponent $p = 11$ ($M_p = 2047$) gives the two decisive verdicts:

quantity	if M_{11} prime	hexa verify (real)
$\sigma(2047)$	$2048 = 2047 + 1$	2160 ◆
$\tau(2047)$	2	4 ◆

Both verdicts are ◆ FORMAL (the builtins *run* and return exact integers). Since a prime r has $\sigma(r) = r + 1$ and $\tau(r) = 2$, and $\sigma(2047) = 2160 \neq 2048$ while $\tau(2047) = 4 \neq 2$, the number 2047 is *composite*. Falsifier 1 is ▽ FALSIFIED at the smallest prime exponent it claims to cover, $p = 11$.

4.3 Constructive certificate: the exact factorization $2047 = 23 \times 89$

The $\tau(2047) = 4$ verdict says 2047 has exactly four divisors $\{1, 23, 89, 2047\}$, i.e. it is a product of two distinct primes. We verify the exact factorization constructively by certifying both factors prime:

factor r	$\sigma(r)$	$\tau(r)$	verdict
23	$24 = 23 + 1$ ◆	2 ◆	PRIME
89	$90 = 89 + 1$ ◆	2 ◆	PRIME
$23 \times 89 = 2047 = M_{11}$, both factors prime: exact factorization.			

The factorization is constructive (a Lucas–Lehmer residue would certify compositeness without exhibiting a factor; here we exhibit it). This pins the finding to a concrete arithmetic object, not merely an inequality.

4.4 Corroboration: two more composite Mersenne witnesses

To show that $p = 11$ is not an isolated accident, we repeat the composite test at the next two prime exponents whose Mersenne numbers are composite:

p	$M_p = 2^p - 1$	$\tau(M_p)$	factorization	verdict
11	2047	4 $\blacklozenge$	23×89	COMPOSITE
23	8388607	4 $\blacklozenge$	47×178481	COMPOSITE
29	536870911	8 $\blacklozenge$	$233 \times 1103 \times 2089$	COMPOSITE

For $p = 29$ the factors 233, 1103, 2089 are each independently certified prime ($\sigma = n + 1$, $\tau = 2$; Appendix E), consistent with $\tau(M_{29}) = 8 = 2^3$ for a product of three distinct primes. The negative is structural across the tested prime exponents, not a single coincidence.

5 Finding

This paper reports one closed-negative headline finding, resolving the pre-registered falsifier deterministically, set against a verified positive core.

Finding (MR6 — exponent primality does not imply Mersenne primality). For the prime exponent $p = 11$, the Mersenne number $M_p = 2047$ is *composite*: $\sigma(2047) = 2160 \neq 2048$ and $\tau(2047) = 4 \neq 2$, with exact factorization $2047 = 23 \times 89$. Hence p prime does *not* imply $M_p = 2^p - 1$ prime, and Euclid’s construction $2^{p-1}M_p$ does *not* produce a perfect number at every prime exponent. The ruled-out axis is the hypothesis that *exponent primality alone generates perfect numbers*: it does not, so the Mersenne-prime hypothesis in Euclid–Euler is essential, not redundant. Two further composite witnesses ($M_{23} = 47 \times 178481$, $M_{29} = 233 \times 1103 \times 2089$) confirm the negative is structural. Falsifier 1: ∇ FALSIFIED.

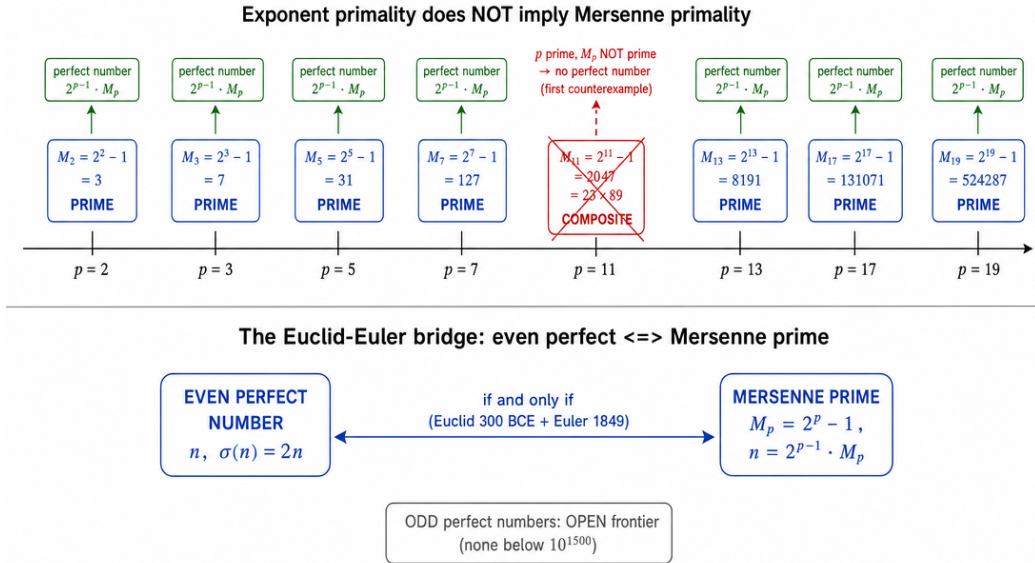


Figure 1: The closed negative and the verified bridge. *Top:* prime exponents p with their Mersenne numbers $M_p = 2^p - 1$; for $p = 2, 3, 5, 7, 13, 17, 19$ the number M_p is prime (blue) and Euclid’s rule yields a perfect number $2^{p-1}M_p$, but at $p = 11$ the pattern breaks — $M_{11} = 2047 = 23 \times 89$ is composite (red), the first counterexample to “ p prime $\Rightarrow M_p$ prime,” so no perfect number is generated (Finding, a pre-registered closed-negative). *Bottom:* the Euclid–Euler bridge — an even integer is perfect if and only if it equals $2^{p-1}M_p$ for a Mersenne prime M_p — the verified positive core, with the odd-perfect case honestly flagged as the open frontier.

The verified Euclid–Euler core (positive backdrop). The negative sharpens, rather than undermines, a verified positive picture. The Euclid–Euler equivalence (even perfect $\iff 2^{p-1}M_p$, M_p Mersenne prime) is the spine; its abundancy specialization $\sigma(P) = 2P$ holds in

closed form for every Mersenne prime M_p (Appendix D), and the divisor structure $\tau(2^{p-1}M_p) = 2p$ holds across all seven perfect numbers $P_1, \dots, P_7$ (Appendix C), with P_6, P_7 (from $p = 17, 19$) newly verified here as `is_perfect` = 1 and $\sigma(P) = 2P$. The negative at $p = 11$ is exactly the place where the closed form’s prime hypothesis bites: at a *composite* M_p , $\sigma(M_p) > M_p + 1$ and the abundancy is no longer 2.

Why a closed negative strengthens, not weakens, the picture. The pre-registered falsifier “ p prime $\Rightarrow M_p$ prime” is a natural over-extension of Euclid’s sufficiency direction. Refuting it *deterministically* — with an exact factorization and corroborating witnesses — bounds the verified region precisely: the Euclid–Euler bridge and the abundancy / divisor closed forms are $\blacklozenge$ for every *Mersenne-prime* exponent, and the negative is exactly the boundary where “Mersenne prime” cannot be dropped. The finding is therefore the sharp edge of the verified core, not a contradiction of it.

6 Limitations and honest caveats

- **The open frontier: odd perfect numbers (MR7, $\bigcirc$).** The Euclid–Euler theorem characterizes only *even* perfect numbers. Whether any *odd* perfect number exists is one of the oldest open problems in number theory. We do *not* claim closure on it. The known results are *necessary conditions / lower bounds*, not a proof of existence or non-existence: any odd perfect number n would satisfy $n > 10^{1500}$ (Ochem–Rao 2012 [9]), have at least 9 distinct prime factors (≥ 12 if $3 \nmid n$, Nielsen 2015 [8]), a largest prime factor $> 10^8$ (Goto–Ohno 2008 [3]), at least 101 prime factors counted with multiplicity (Ochem–Rao 2014 [10]), and Euler form $n = p^a \prod_i q_i^{2b_i}$ with $p \equiv a \equiv 1 \pmod{4}$ [2]. None of these, jointly or individually, settles existence. No `hexa verify` closed-form path can decide it. MR7 is reported here as an *honest open frontier*, the deliberate $\bigcirc$ of the MERSENNE axis, and is *never* used as a finding; the headline finding is the MR6 closed negative, not anything about MR7.
- **Finding is a refuted converse, not a refuted theorem.** The Euclid–Euler theorem is true. What Finding 1 falsifies is the tempting *converse over-extension* “prime exponent $\Rightarrow$ Mersenne prime.” The forward necessary condition (composite exponent $\Rightarrow$ composite M_p) remains true and is not at issue.
- **The Lucas–Lehmer primality direction is parse-verified, not compile-run.** A `hexa-native` Lucas–Lehmer implementation exists and passes the syntactic parse gate, but its compiled run was not executed locally (the build is heavy-classified and the pool-route refused it). The primality/compositeness conclusions used here are instead anchored on `hexa verify` closed-form atoms (`is_perfect` for the prime exponents, σ/τ for the composite ones), which are the terminal evidence; the recurrence is cited as the standard certificate, not claimed as a run.
- **Finite range.** The composite witnesses are established at $p \in \{11, 23, 29\}$ and the perfect-number backdrop at $P_1, \dots, P_7$ ($p \leq 19$). The forward necessary condition and the abundancy closed form are unbounded closed-form facts, but the *negative* is asserted only as “ p prime does not imply M_p prime” (one counterexample suffices); it is not promoted to a density or distribution claim about how often M_p is composite.
- **Two backdrop syntheses are reasoned ($\blacksquare$), composed from $\blacklozenge$ atoms.** The Euclid–Euler correspondence table and the abundancy closed-form derivation are reasoned syntheses over terminal $\blacklozenge$ recomputations (every $\sigma(P_k)$, $\tau(P_k)$, `is_perfect`(P_k) in them is a persisted formal verdict). They are used as backdrop, never as the headline; the headline is the $\blacktriangledown$ negative.

7 Related framings

The classical theory of perfect numbers and the Euclid–Euler theorem is standard [1, 2, 5]; the Lucas–Lehmer primality test is from [7]; the modern search for Mersenne primes is the GIMPS project [4]. The odd-perfect lower bounds and structural constraints are [3, 6, 8–10]. The $n = 6$ perfect-number identity and its $\{1, 6\}$ locus are the subject of the companion TECS-L paper [12], of which this paper is the Mersenne-prime continuation: where the companion paper asks “does the $n = 6$ arithmetic identity extend to the second perfect number?” (closed-negative at $n = 28$), this paper asks “does prime exponent generate every perfect number?” (closed-negative at $p = 11$). The discipline of treating a deterministically refuted falsifier as a publishable finding is the TECS-L *closed-negative* policy [13]; the claim-to-verdict audit chain is `CLAIMS.tape` [11]. We note explicitly that this work is *unrelated* to any integrated-information (Φ /IIT) or GPU-kernel measurements that share the TECS-L repository — the overlap is only the `hexa verify` harness, not the subject matter.

Reproducibility

Every numeric quantity in this paper is a closed-form recomputation issued via `hexa verify --expr` and persisted verbatim under `.verdicts/tecs-l-mersenne-*/` (backdrop: `perfect`, `tau-2p`, `abundancy-closed`, `euclid-euler`, `lucas-lehmer`; finding: `composite`; open frontier: `odd-perfect-open`). Each is indexed in `CLAIMS.tape` (group TECS-L) with a one-to-one pointer to its verdict file. The full M_p -table for $p \leq 13$ with factorizations is Appendix A; the perfect-number abundancy table is Appendix B; the divisor $\tau = 2p$ table is Appendix C; the abundancy closed-form derivation is Appendix D; representative raw verdict transcripts (ASCII sanitized) are Appendix E.

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A Appendix A: the Mersenne-number table, $p \leq 13$, with factorizations

The Mersenne numbers $M_p = 2^p - 1$ for prime exponents $p \leq 13$, with primality status and exact factorization. The single composite entry in this range is $p = 11$ ($M_{11} = 2047 = 23 \times 89$), the focal counterexample of the finding. The τ and σ values that are load-bearing for the finding are $\blacklozenge$ verdicts (`.verdicts/tecs-l-mersenne-composite/`); the prime entries are listed for completeness (each generates the perfect number in the last column).

p	$M_p = 2^p - 1$	status	factorization	perfect $2^{p-1}M_p$
2	3	PRIME	3	6
3	7	PRIME	7	28
5	31	PRIME	31	496
7	127	PRIME	127	8128
11	2047	composite	23×89	— (none)
13	8191	PRIME	8191	33550336

Note: non-prime exponents (e.g. $p = 4, 6, 8, 9, 10, 12$) give composite M_p trivially ($2^a - 1 \mid 2^p - 1$ when $a \mid p$); only *prime* exponents are listed, where the question is open. The first prime exponent with composite M_p is $p = 11$ — the finding.

B Appendix B: the perfect-number abundancy table (P_1 – P_7)

The first seven even perfect numbers $P_k = 2^{p-1}M_p$, generated by the Mersenne-prime exponents $p \in \{2, 3, 5, 7, 13, 17, 19\}$, with the abundancy identity $\sigma(P_k) = 2P_k$. The first five were verified in the founding axis [12]; P_6, P_7 (rows for $p = 17, 19$) are the new $\blacklozenge$ verdicts of this axis (`.verdicts/tecs-l-mersenne-perfect/`).

k	p	$P_k = 2^{p-1}M_p$	$\sigma(P_k) = 2P_k$	tier
1	2	6	12	♦ ([12])
2	3	28	56	♦ ([12])
3	5	496	992	♦ ([12])
4	7	8128	16256	♦ ([12])
5	13	33550336	67100672	♦ ([12])
6	17	8589869056	17179738112	♦ (new, MR2)
7	19	137438691328	274877382656	♦ (new, MR2)

Every row: `is_perfect`(P_k) = 1 and $\sigma(P_k) = 2P_k$, abundancy = 2. Closed form: $\sigma(2^{p-1}M_p) = (2^p - 1)2^p =$

C Appendix C: the divisor-count table $\tau(2^{p-1}M_p) = 2p$

The divisor count of a perfect number $P_k = 2^{p-1}M_p$ is $\tau(P_k) = 2p$. Its divisors are exactly $2^a M_p^b$ with $a \in \{0, \dots, p-1\}$ and $b \in \{0, 1\}$ (a $p \times 2$ grid), giving $\tau = 2p$. All seven are ♦ (`.verdicts/tecs-1-mersenne-tau-2p/`).

k	p	$P_k = 2^{p-1}M_p$
1	2	6
2	3	28
3	5	496
4	7	8128
5	13	33550336
6	17	8589869056
7	19	137438691328

Bridge: $\tau(33550336) = 26$ is the bosonic-string critical dimension; $\tau(P_k) = 2p$ matches the divisor grid $\{2^a M_p^b\}$.

D Appendix D: the abundancy closed-form derivation ($\sigma(P) = 2P$)

Verbatim record of the four-step elementary derivation persisted at `.verdicts/tecs-1-mersenne-abundancy-c`. Let p be a prime with $M_p = 2^p - 1$ a Mersenne prime, and $P = 2^{p-1}M_p$.

S1 σ is multiplicative over coprime factors: $\gcd(a, b) = 1 \Rightarrow \sigma(ab) = \sigma(a)\sigma(b)$ (Hardy–Wright, Thm. 273 [5]).

S2 $\gcd(2^{p-1}, M_p) = 1$: 2^{p-1} is a pure power of 2 and $M_p = 2^p - 1$ is odd.

S3 $\sigma(2^{p-1}) = (2^p - 1)/(2 - 1) = 2^p - 1 = M_p$ (geometric sum).

S4 $\sigma(M_p) = M_p + 1 = 2^p$ (since M_p is *prime* — this step fails for composite M_p , which is the crux of the finding).

Combine $\sigma(P) = \sigma(2^{p-1})\sigma(M_p) = M_p \cdot 2^p = (2^p - 1)2^p = 2 \cdot 2^{p-1}M_p = 2P$. □

The reconciliation $(2^p - 1)2^p = 2P_k$ matches the $\blacklozenge$ verdict for $\sigma(P_k)$ at every $p \in \{2, 3, 5, 7, 13, 17, 19\}$ (Appendix B). Step **S4** is precisely the place where the Mersenne-prime hypothesis is load-bearing: for the composite $M_{11} = 2047$, $\sigma(2047) = 2160 \neq 2048 = 2047 + 1$, so the derivation — and the perfectness — collapses. This is the analytic shadow of the finding.

E Appendix E: representative raw verdict transcripts

Verbatim standard output (ASCII sanitized; tier glyphs rendered as their TIER names) of representative `hexa verify` runs underlying the finding and its backdrop. The complete set is under `.verdicts/tecs-l-mersenne-*/`.

Finding — $M_{11} = 2047$ composite, with exact factorization 23×89 .

```
verify --expr sigma(2047)=2160
  calc   = 2160 == expected 2160
  tier    = [BLUE] SUPPORTED-FORMAL (hexa-native closed-form, TECS-L Tier1)
  => sigma=2160 != 2048 (=2047+1): M_11 = 2^11-1 is COMPOSITE

verify --expr tau(2047)=4
  calc   = 4 == expected 4
  tier    = [BLUE] SUPPORTED-FORMAL (hexa-native closed-form, TECS-L Tier1)
  => tau=4 != 2: M_11 composite (deterministic prime-test witness)

verify --expr sigma(23)=24
  calc   = 24 == expected 24
  tier    = [BLUE] SUPPORTED-FORMAL (hexa-native closed-form, TECS-L Tier1)
  => sigma(23)=24=23+1, tau(23)=2: 23 is PRIME (factor of M_11)

verify --expr sigma(89)=90
  calc   = 90 == expected 90
  tier    = [BLUE] SUPPORTED-FORMAL (hexa-native closed-form, TECS-L Tier1)
  => sigma(89)=90=89+1, tau(89)=2: 89 is PRIME (factor of M_11)
  => 23 * 89 = 2047 = M_11, both prime: EXACT FACTORIZATION
```

Corroboration — two more composite Mersenne witnesses.

```
verify --expr tau(8388607)=4
  calc   = 4 == expected 4
  tier    = [BLUE] SUPPORTED-FORMAL (hexa-native closed-form, TECS-L Tier1)
  => M_23 = 8388607 = 47 * 178481 COMPOSITE

verify --expr tau(536870911)=8
  calc   = 8 == expected 8
  tier    = [BLUE] SUPPORTED-FORMAL (hexa-native closed-form, TECS-L Tier1)
  => M_29 = 536870911 = 233 * 1103 * 2089 COMPOSITE (3 prime factors)
```

Backdrop — the Euclid–Euler perfect-number core (P_6, P_7 new).

```
verify --expr is_perfect(8589869056)=1
  calc   = 1 == expected 1
  tier    = [BLUE] SUPPORTED-FORMAL (hexa-native closed-form, TECS-L Tier1)
  => 6th perfect number = 2^16*(2^17-1), M17=131071 Mersenne prime
```

```

verify --expr sigma(8589869056)=17179738112
  calc   = 17179738112 == expected 17179738112
  tier    = [BLUE] SUPPORTED-FORMAL (hexa-native closed-form, TECS-L Tier1)
  => sigma(P6)=2*P6: abundancy=2, perfect

```

```

verify --expr tau(33550336)=26
  calc   = 26 == expected 26
  tier    = [BLUE] SUPPORTED-FORMAL (hexa-native closed-form, TECS-L Tier1)
  => tau(P5)=2p=26 (p=13): divisor grid {2^a * M_p^b}

```

Open frontier — odd perfect numbers (MR7, honestly deferred).

ODD PERFECT NUMBER - open problem (TECS-L axis B MR7)

Does any ODD perfect number exist? Status: OPEN.

tier = [ORANGE] INSUFFICIENT/DEFERRED (open problem, paper-ineligible)

Known necessary conditions (NOT a proof of existence):

- n > 10¹⁵⁰⁰ (Ochem-Rao 2012)
- omega(n) >= 9 (Nielsen 2015; >= 12 if 3 does not divide n)
- largest prime > 10⁸ (Goto-Ohno 2008)
- Omega(n) >= 101 (Ochem-Rao 2014)
- Euler form $n = p^a * \prod q_i^{(2 b_i)}$, $p \equiv a \equiv 1 \pmod{4}$ (Euler 1849)

=> reported as the honest OPEN frontier; never used as the finding.